

1–2. Complete systems of invariants for the ternary biquadratic form

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The problem is the construction of the complete system of concomitants belonging to the ternary quartic form. I shall use the symbolic method, because it keeps visible the two operations that are decisive: polarization and contraction. Let

$$f = a_x^4 = b_x^4 = c_x^4 = \dots$$

be the symbolic notation for the ground form. The symbolic letters are not independent coefficients, but devices for writing invariant differential operations. An expression built from determinants such as

$$(abc) = \det(a, b, c)$$

and from factors such as a_x, b_x, c_x is to be read as a covariant after umbral substitution. The task is therefore not merely to list expressions, but to determine a finite generating set from which every symbolic expression of the correct total degree follows by algebraic combination.

The ternary biquadratic is the first case in which the older methods of transvectants cease to be short. The number of possible symbolic products grows rapidly, and many apparent forms are reducible. One must distinguish at every stage between a newly appearing form and a form obtained from lower ones by multiplication, by applying the Aronhold operator, or by taking a polar. The guiding principle is that every concomitant can be assigned a type: its degree in the coefficients, its order in the variables, and its class in the dual variables. Forms of different type cannot cancel. This permits the computation to be organized by tables rather than by trial.

I use the following elementary reductions repeatedly. If a symbolic product contains two equal symbolic letters in a determinant, the product vanishes. If the same determinant occurs

with its letters permuted, only the sign changes. If a symbolic letter occurs to a power higher than the degree allowed by the ground form, the expression can be replaced by a linear combination with lower powers. Finally the fundamental identities among determinants in three letters, obtained by expanding a four-by-four determinant with two identical rows, give all linear syzygies needed for the reduction of the symbolic monomials.

For example, an expression of the form

$$(abc)^2 a_x^2 b_x^2 c_x^2$$

is not treated as an isolated object. It is compared with all products obtained by replacing one occurrence of x by a polar variable, by contracting two letters, and by multiplying invariants of smaller degree. If after all these operations no reduction is possible, the expression is entered as a member of the fundamental system. If a reduction is possible, it is recorded in the table of dependencies. The same procedure is then repeated at the next symbolic degree.

The computation naturally divides into invariants, covariants, contravariants, and mixed concomitants. The invariants have no remaining variables. The covariants contain only point variables; the contravariants contain only line variables; the mixed forms contain both. The passage from covariants to contravariants is performed by the dual operation, but one must not suppose that this simply halves the work. Some forms are self-dual only after multiplication by a determinant factor, and some reductions have no visible dual until they are written in normalized symbolic form.

The Aronhold process is used in the following way. Suppose that a covariant of order r has been found. Applying the invariant differential operator attached to a second symbolic copy of the ground form produces a new covariant of order $r + 4 - 2s$, where s is the number of contractions performed. Terms that contain a factor already known to be reducible are immediately discarded. The remaining terms are compared against the current table. In this manner all forms of a given degree are obtained from forms of smaller degree together with a finite list of primitive exceptions.

The result of the calculation is a finite complete system. It is important that the word complete be understood in the strict sense: every invariant and every concomitant of the ternary quartic is an integral rational polynomial in the listed forms. There is no appeal to analytic continuity or to a generic specialization. The proof consists of the symbolic reduction itself, together with the verification that the reductions are compatible with specialization of the coefficients. Thus the system is not a system only for the general quartic, but for every ternary quartic.

The reductions can be arranged according to weight. Let $M_{d,r,s}$ denote the module of symbolic concomitants of coefficient degree d , point order r , and line order s . The products of earlier members span a submodule $P_{d,r,s}$. What has to be found is a basis of a complement of $P_{d,r,s}$ in $M_{d,r,s}$. The symbolic identities give a presentation of $M_{d,r,s}$; hence the computation is finite. In the cases where the quotient is zero, no new form occurs. Where it is non-zero, the number of new fundamental forms is exactly its rank.

This also explains why the construction is independent of the symbolic notation. If one writes the quartic as a polynomial in coefficients a_{ijk} , the concomitants become polynomials in these coefficients and in the auxiliary variables. The symbolic calculus is simply a compressed notation for these polynomials. Every reduction identity is an identity in the coefficient ring. Thus the final system is a system in the ordinary algebraic sense.

The decisive point of the dissertation is therefore not one particular invariant, but the method of closing the table. After a certain degree the Aronhold operations and products account for every symbolic type. This is checked by the dimension count supplied by the Cayley series. The coefficient of the corresponding term in the series gives the number of linearly independent concomitants of that type. The table of already generated products reaches this number; consequently there are no missing primitive forms.

In this form the computation is also a test of the general finiteness philosophy in invariant theory. The individual formulae are long, and their printed arrangement is necessarily compressed. But the proof does not rely on a guess that the list is complete. Completeness

follows from the comparison between the symbolic reduction and the enumerative series. The list is therefore not a catalogue in the empirical sense, but the outcome of a finite algorithm. I shall use in the sequel the following convention for the tables. A new form is marked by its symbolic representative, followed by its degree and its order-class pair. If two representatives differ only by a determinant identity, only one is retained. If a form is obtained by multiplication, it is written in the product column and not repeated as a primitive form. If a form arises from a transvection of lower forms, the generating operation is recorded. This convention makes the tables reproducible and prevents the same concomitant from being counted twice.

The complete system contains forms of several quite different geometric meanings. Some express special contacts of the quartic with lines, others express the presence of singular polar curves, and still others measure degenerations of the net of conics associated with the quartic. The invariant-theoretic calculation itself does not require these interpretations. Nevertheless they provide a useful check: when the symbolic form is specialized to a quartic with a known singularity, the corresponding invariant must vanish with the predicted multiplicity.

The final theorem can be stated briefly. The concomitants displayed in the tables generate the full algebra of concomitants of the ternary quartic. Every concomitant of the ground form is a polynomial expression in them. Moreover the stated symbolic relations suffice to reduce any symbolic expression to such a polynomial. The proof is by induction on coefficient degree, using the reduction lemmas and the dimension comparison at each bidegree. This completes the construction of the form system.

Additional reductions and formulation

The enumeration is best read as an induction on the coefficient degree. In degree one there is only the ground form and its polar relatives. In degree two the determinant factors first appear, and the possible concomitants are still few enough to list directly. In higher degrees the number of symbolic monomials is large, but all monomials have a finite signature: the

number of determinant brackets, the remaining powers of each symbolic letter, and the distribution of point and line variables. The reduction preserves this signature except when it replaces a product by forms of lower degree. This is why the induction terminates.

At several places the calculation uses what may be called the exchange of letters. A symbolic product with letters a, b, c, d may be expanded by a determinant identity and rewritten as a sum in which one of the letters occurs in a simpler position. Repeating the exchange eventually brings every product to a normal arrangement. The signs introduced by the exchanges are fixed by the orientation of the determinants. Once a normal arrangement is chosen, two symbolic products can be compared unambiguously.

The Cayley series supplies an independent verification of the enumeration. For each type (d, r, s) it gives the number of independent concomitants that can exist. The table of products generated from earlier forms gives a lower bound on that number, and the symbolic reductions give an upper bound. When the two bounds agree, the entry is closed. When they do not agree, the difference is exactly the number of new primitive forms to be inserted. This is the numerical mechanism behind the phrase complete system.

A point that is easy to miss in the printed tables is that the same concomitant can occur in several symbolic disguises. For instance, a product obtained by first polarizing and then contracting can coincide with one obtained by first multiplying and then applying a transvection. The dissertation resolves this by retaining one representative and recording the identity that identifies the others with it. Thus the table is not only a list of generators; it is also a record of the reductions among candidate generators.

The specialization argument is also essential. A symbolic identity proved for a general quartic is an identity in the coefficients and therefore remains true for a singular quartic. Conversely, the independence count is made in the generic case; if a generator specializes to zero for a special quartic, this does not remove it from the complete system. The system is an algebraic system over the coefficient ring, not a minimal system for each individual member of the family.

The complete system therefore has two roles. It gives a finite algebra of concomitants for calculations with a fixed ternary quartic, and it gives a coordinate algebra for the quotient problem in which quartics are considered up to projective transformation. The second role explains why it is not enough to know invariants alone. Covariants and mixed concomitants encode the geometry of associated points, lines, and polar curves; they are needed to reconstruct the full projective information.

Further proof details

A final check concerns relations among the fundamental forms themselves. The theorem that a list generates all concomitants is distinct from the problem of finding all syzygies among the generators. The dissertation uses only those syzygies needed to prove generation. Some relations are displayed because they are required for reductions; others are consequences of the symbolic determinant identities but are not needed for the proof of completeness. Thus the printed system is a generating system, not a free system.

The low-degree entries also serve as anchors for the induction. If an alleged reduction in higher degree uses a product of lower forms, each of those lower forms must already have been normalized. Otherwise the same expression could be counted under two different names. This is why the tables proceed in a fixed order and why every operation is recorded at the point where it first becomes available.

Geometrically, the ternary quartic is a plane curve of degree four. Its invariants detect projective features of the curve; its covariants produce associated curves and point configurations. The symbolic system therefore encodes both the algebra of the coefficients and the projective geometry of the curve. The computation is long because these two aspects are interwoven, not because the principles are obscure.

5. Rational function fields

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Let K be a rational function field over a ground field k . The question considered here is how the property of being rational can be recognized without reference to a special system of independent variables. If

$$K = k(x_1, \dots, x_n),$$

then every other system of algebraically independent generators gives the same field. The problem is to determine which subfields and extensions obtained from such a field are again rational and how this can be detected by intrinsic algebraic operations.

A first observation is elementary but useful. If $u_1, \dots, u_n$ are algebraically independent and K is algebraic over $k(u_1, \dots, u_n)$, then equality need not hold. Rationality is therefore stronger than having the correct transcendence degree. One must ask for a transformation whose inverse is again rational. Thus the central object is the group of birational substitutions of the variables, or equivalently the automorphism group of $k(x_1, \dots, x_n)$ over k .

In one variable the theorem is classical: every subfield of $k(x)$ of transcendence degree one is again a rational function field. This is the algebraic form of the theorem that the projective line has genus zero. In several variables the analogous statement is false in this simple form. Nevertheless many subfields arising from linear substitutions or from finite groups of substitutions remain rational, and the proof can often be reduced to a choice of suitable invariants.

Suppose first that a finite group G acts linearly on $k(x_1, \dots, x_n)$, permuting or linearly transforming the variables. The fixed field

$$K^G = \{z \in K : z^S = z \text{ for all } S \in G\}$$

has the same importance for rational function fields that invariant rings have for polynomial forms. The rationality question asks whether K^G can be written as $k(y_1, \dots, y_n)$. In favorable cases the elementary symmetric functions, or their analogues for the given representation, give such a system.

The method used here is to separate the algebraic and the transcendental parts of the argument. Choose a separating transcendence basis of the fixed field. Then the original field is algebraic over this fixed field; its degree is bounded by the order of the group. If one can show that the chosen invariants already give an extension of degree equal to the group order, then they form a full rational system. Thus a field-theoretic degree calculation replaces the direct construction of an inverse substitution.

For permutation groups this gives the familiar theorem immediately. If G is the full symmetric group on the variables, the elementary symmetric functions generate the fixed field. For subgroups the situation is subtler. One may need resolvent functions whose conjugates under the symmetric group separate the cosets of G . The corresponding fixed field is then generated by the symmetric functions together with one resolvent root. Rationality is obtained only when this root can be replaced by a rational parameter system.

The note emphasizes that rationality is invariant under extension of the ground field only in one direction. If a field becomes rational after adjoining constants, it does not necessarily follow that it was rational over the original field. The obstruction is a descent obstruction. In the language used later, it is expressed by the failure of a system of conjugate generators to be definable over the smaller field. Even when the algebraic relations have coefficients in k , the generators themselves may be defined only over an extension.

This distinction is important for invariant theory. A complete system over an algebraically closed field may descend only after replacing its generators by invariant combinations. Conversely, rational generators over a splitting field may satisfy conjugacy relations that prevent them from giving rational generators over k . Thus the rationality of the field of invariants is not a purely geometric question; it also involves arithmetic of the constants.

The guiding lemma is the following. Let L/K be a finite Galois extension and let x be transcendental over L . If the group acts on $L(x)$ by fractional linear transformations in x , then the fixed field is rational over the fixed field of L , provided that a fixed divisor of degree one exists. In elementary terms, one constructs a parameter by taking a quotient of two semi-invariant linear forms. The proof is just the transformation law for binary linear forms, rewritten in field language.

The conclusion is not a classification theorem, but a collection of reductions. To decide whether a field is rational one should first isolate the finite algebraic part, then test whether the remaining transcendental variables can be chosen invariantly. This method will be used again in the more systematic treatment of fields and systems of rational functions. The present note records the field-theoretic core of that treatment.

Additional reductions and formulation

The one-variable theorem may be used as a model for the general method. If F is a subfield of $k(x)$ of transcendence degree one, choose an element $u \in F$ of minimal degree as a rational function of x . Any other element of F is algebraic over $k(u)$, but the minimality forces the degree of $k(x)$ over $k(u)$ to coincide with the degree over F . Hence $F = k(u)$. In higher dimension no such simple minimal degree argument exists, and the failure of this argument is precisely where the new difficulties begin.

For finite groups the fixed field can be approached through the normal basis idea. A generic element z has conjugates z^S under the group. The elementary symmetric functions of these conjugates lie in the fixed field, and z satisfies the corresponding equation. If the conjugates are all distinct, the degree of the equation is the group order. The fixed field is then obtained by adjoining to the symmetric fixed field enough coefficients to describe the orbit. This is the field version of the resolvent construction.

The rationality of a fixed field is often proved by finding triangular generators. One first chooses an invariant involving all variables, then replaces one variable by a quotient invariant,

and proceeds by induction. In permutation examples this produces elementary symmetric functions. In affine examples it produces differences and sums. The important point is that each step must be reversible by rational operations; otherwise one has only algebraic generation, not rational generation.

Descent over the ground field can be stated concretely. Suppose L/k is a finite extension and $L \otimes_k F$ is rational over L . Choose rational generators y_i . For each automorphism S of L/k , the elements $S(y_i)$ form another rational generating system. Descent asks whether there is a system of rational functions in the y_i , invariant under all S , that still generates the same field. This is generally a non-trivial problem of simultaneous rational transformation.

Thus even the modest question whether a field is rational leads to three layers: algebraic dependence, birational inversion, and descent of constants. The first is decided by prime ideals of relations. The second is decided by the degree of the field over a chosen transcendence basis. The third is decided by the Galois action on possible bases. Keeping these layers separate prevents several common mistakes in the use of rational parameters.

Further proof details

One practical test for rationality is the construction of a Lüroth-type parameter on a divisor or pencil. If a family of hypersurfaces has a rational member through the general point and the residual intersection is controlled, the quotient of two equations of the pencil may serve as a parameter. This geometric argument translates into the field statement that a subfield generated by a pencil has transcendence degree one and is therefore rational.

In several variables, the failure of rationality is often invisible if one looks only at general points. A unirational field can be covered by rational parameters without being rational itself. Field-theoretically this means that the ambient rational field is finite over the subfield, but the degree is greater than one. The rational functions give a dominant parametrization; they do not give a birational inverse. This distinction is one of the motivations for the later systematic use of field degrees.

The note's conclusions are deliberately modest. It does not claim that every fixed field of a finite group is rational. Rather, it isolates the constructions that do give rational fixed fields and indicates where the obstruction lies. This formulation later became one of the standard forms of Noether's problem: decide whether the field of invariants of a rational field under a finite group is again rational over the ground field.

6. Fields and systems of rational functions

Original publication: Mathematische Annalen 76 (1915), pp. 161–196.

Let K be a field of rational functions in finitely many variables over a ground field k . A system of rational functions

$$f_1, \dots, f_m \in K$$

defines a subfield $k(f_1, \dots, f_m)$. The basic problem is to determine when two systems define the same subfield, when one system can be transformed into another by rational substitutions, and when a subfield so obtained is itself purely transcendental. These questions are the natural field-theoretic counterpart of elimination theory.

The first step is to replace a system of functions by the ideal of algebraic relations among them. If $Y_1, \dots, Y_m$ are indeterminates, the kernel of the homomorphism

$$k[Y_1, \dots, Y_m] \longrightarrow K, \quad Y_i \mapsto f_i,$$

is a prime ideal. Its quotient field is the field $k(f_1, \dots, f_m)$. Thus the study of systems of rational functions is equivalent to the study of prime ideals together with their quotient fields. This formulation makes possible the use of dimension and specialization arguments.

A subsystem $f_{i_1}, \dots, f_{i_r}$ is called independent if there is no algebraic relation among its members. A maximal independent subsystem has cardinality equal to the transcendence degree of the generated field. The remaining functions are algebraic over the field generated

by such a subsystem. However, different maximal independent subsystems need not give isomorphic algebraic extensions. The dependence on the chosen basis is controlled by the field degree of the total system over that basis.

The main technical device is normalization by a linear change of variables. After replacing the functions by general linear combinations, one may assume that the first r functions form a separating system and that the remaining functions are integral over the polynomial ring generated by them, after clearing denominators. This is a rational-function version of Noether normalization. It permits all later arguments to be carried out over a polynomial subring, where denominators and exceptional specializations can be controlled.

Let $F = k(f_1, \dots, f_m)$. A rational representation of F is a choice of algebraically independent elements $u_1, \dots, u_r$ such that F is algebraic over $k(u_1, \dots, u_r)$. The degree of this algebraic extension is the order of the representation. If the order is one, the representation is parametric: the field is rational. The paper shows how to test the order by eliminating the original variables and computing the degree of the resulting prime ideal over the chosen parameters.

The role of resultants is purely auxiliary. A resultant may eliminate variables and produce an equation for one function over the others, but the resultant often contains extraneous factors introduced by denominators or by components at infinity. The correct object is the prime ideal of relations. Resultants must be saturated with respect to the product of the denominators and then decomposed. Only the component corresponding to the given field is relevant.

The same point is visible in geometric language. The rational functions define a rational map from affine or projective space into another space. The field F is the function field of the image. Exceptional divisors and base loci may alter equations in the image if they are not removed. Working with prime ideals and quotient fields suppresses these accidental components. Birational equivalence is then equality of function fields, not equality of one particular set of equations.

A useful criterion is obtained from specialization. Suppose the coefficients of the functions

depend on parameters. If for one specialization the functions are algebraically independent and the relevant degree is d , then outside a proper exceptional set the same independence and the same degree hold. This is because the leading coefficients and discriminants that control the integral equations vanish only on a proper algebraic subset. Thus generic rationality questions can be reduced to one well-chosen specialization.

The treatment also separates absolute and relative rationality. A field may be rational over an algebraic extension of k but not over k itself. If rational generators over the larger field are permuted by the Galois group, the descent question becomes the problem of finding invariant combinations that remain algebraically independent and generate the same field. This is the same obstruction encountered in the theory of rational function fields and will later reappear in invariant theory and Galois theory.

The principal theorem can be formulated as follows. Every field generated by a finite system of rational functions admits a finite normal representation over a purely transcendental subfield; after a general linear transformation the generators may be chosen among linear combinations of the original functions. Equality of two such fields is decided by equality of the corresponding prime ideals after passage to quotient fields. Rationality is the special case in which the normal representation has degree one.

In applications, one begins with explicit functions and forms the relation ideal. One then chooses a maximal independent subsystem and computes the degree of the remaining algebraic extension. If the degree is one, the system is rational and the chosen subsystem is a rational parameter system. If the degree is larger, no rational inverse exists for that particular subsystem; one may still try another subsystem or a birational transformation, but the obstruction is now a genuine field degree.

The significance of the paper lies in the replacement of formula manipulation by field-theoretic invariants. A rational system is not an array of formulae but a finitely generated field extension. Its essential data are transcendence degree, algebraic degree over a normalizing subfield, and the prime ideal of relations. These data survive changes of variables and make the theory

independent of accidental representations.

Additional reductions and formulation

The proof of normalization is elementary in spirit. Let the functions be written with common denominator, and let the numerator variables be subjected to a general linear transformation. The bad transformations are those for which some leading coefficient vanishes or some algebraic dependence degenerates. These conditions are algebraic equations in the transformation coefficients. Since they do not hold identically, a transformation outside their zero set has the required property. This is the algebraic content of the word general.

When the relation ideal is prime, the quotient ring still may not be integrally closed. Normalization of the quotient ring corresponds to passing from a singular image variety to its normal model. At the level of function fields this passage changes no rational functions, but it may simplify integral equations. Noether uses this distinction systematically: equations describe a model, whereas the quotient field describes the birational object itself.

Equality of two function fields generated by two systems f_i and g_j can be checked by mutual rational expression. It is enough to show that every f_i is rational in the g_j and every g_j is rational in the f_i . Algebraically this means that the two embeddings of polynomial rings into the ambient field have quotient fields with the same image. The condition is unaffected by clearing denominators, provided that the denominators are not allowed to vanish identically.

The degree of a rational map appears here as a field degree. If the image field F is generated by r independent functions and $[K : F] = d$, then a general point of the image has d preimages in the original parameter space, counted algebraically. Birationality is exactly the case $d = 1$. This gives a purely algebraic criterion for when a system of rational functions can serve as a change of variables.

The paper's method is especially useful when parameters occur. A family of systems defines a prime ideal over a parameter ring after removing exceptional components. The degree over a

normalizing subfield is constant on a dense open set of the parameter space. Hence properties such as independence, separability, and birationality are generic. Special parameter values may be exceptional, but they do not alter the general theorem.

In later ideal-theoretic language one would say that the coordinate ring of the image is a finitely generated domain and that the original rational field is a finite extension of its field of fractions after normalization. The 1915 treatment does not yet use the later vocabulary of primary decomposition in its final form, but it already uses the same strategy: replace computations with equations by computations with prime ideals and quotient fields.

Further proof details

The relation ideal also records implicit equations. Given a rational parametrization, eliminating the parameters gives equations for the image. But the implicit equations alone do not determine whether the parametrization is one-to-one. The field degree over the image field measures this. Thus implicitization and inversion are separate problems: the former belongs to ideals, the latter to quotient fields.

A useful way to organize the calculations is to choose a tower

$$k(u_1, \dots, u_r) \subset F \subset K.$$

The first extension is finite by normalization, and the second is the inclusion of the generated field into the ambient rational field. If $F = K$, the system gives a birational coordinate system. If $F \neq K$, the missing functions represent the algebraic ambiguity of the parametrization. The tower makes the ambiguity explicit.

The paper also anticipates the language of generic freeness. After excluding a finite set of bad denominators, modules that occur in the calculation become free over the parameter ring, and ranks become constant. This justifies specialization arguments: a rank or degree computed at one good specialization is the generic rank or degree. The exceptional cases are

not ignored; they are precisely the zeros of the denominators and discriminants introduced during normalization.

9. The most general domains of completely transcendental numbers

Original publication: 1916.

Consider a domain generated from a ground field by elements subject to no algebraic relations except those forced by the construction. Such elements are called completely transcendental. The question is to describe the most general domains in which these elements can be manipulated as independent quantities while still allowing algebraic adjunctions, specialization, and passage to quotient fields. This is a foundational problem for the algebraic use of parameters.

Let k be a base field and let $x_1, \dots, x_n$ be algebraically independent. The polynomial ring $k[x_1, \dots, x_n]$ is the simplest domain of completely transcendental quantities, and its quotient field is $k(x_1, \dots, x_n)$. But many constructions require adjoining further elements integral or algebraic over this ring, or localizing at systems of denominators. The problem is to determine which properties of independence survive these operations.

A domain R over k is to be viewed through its finite subextensions. Every element of R should lie in a subdomain generated by finitely many transcendental elements and finitely many algebraic elements over them. This finiteness condition prevents pathologies and permits the use of elimination. Thus even a very large domain is controlled by its finitely generated subdomains.

The essential invariant is the number of independent transcendental generators. If R has quotient field K , this number is the transcendence degree of K/k . It is independent of the chosen generators. If $u_1, \dots, u_r$ and $v_1, \dots, v_s$ are two transcendence bases of K/k , then $r = s$.

Moreover each set can be transformed into the other through a finite chain of elementary replacements, at least after restricting to suitable finitely generated subfields. This gives a concrete algebraic meaning to the dimension of the domain.

Specialization is handled by homomorphisms from the polynomial subdomain to algebraic extensions of k . An algebraic relation that holds after all admissible specializations must already hold in the original domain. Conversely, a denominator that is non-zero in the general domain may vanish under exceptional specializations. Therefore one must always exclude a finite union of hypersurfaces when using specialization to prove general statements.

The paper insists on the difference between formal independence and algebraic independence in a fixed field. Formal variables are symbols; algebraically independent elements are elements of an extension field satisfying no polynomial relation. Any identity in formal variables remains true after substitution by algebraically independent elements. But a statement involving denominators or algebraic adjunctions may require restricting the substitutions. This is the reason for introducing domains rather than only fields.

The construction of the most general domain proceeds by taking a universal polynomial domain and then factoring only by the prescribed algebraic relations. If no relations are prescribed, the domain remains a polynomial ring. If algebraic elements are prescribed by equations, one forms the corresponding quotient and then selects the component determined by the desired prime ideal. The quotient field of that component is the most general field satisfying the requirements.

This language clarifies the use of generic points. A generic point of an irreducible variety is not a mysterious geometric object; algebraically it is the homomorphism from the coordinate ring into its quotient field. The coordinates of the generic point are completely transcendental modulo the prime relations defining the variety. Statements about the general point are statements in this quotient field. Exceptional cases correspond to specializations where denominators vanish or algebraic degrees drop.

The final conclusion is that domains of completely transcendental numbers are best treated as

universal domains built from polynomial rings, prime ideals, localizations, and finite algebraic extensions. Their generality is not obtained by allowing infinitely many uncontrolled elements, but by requiring that every finite calculation take place in a finitely generated subdomain with a specified transcendence basis. This is the algebraic setting in which later normalization and ideal-theoretic arguments become possible.

Additional reductions and formulation

A recurring question is whether a construction depends on the names of the transcendental elements. It should not. If $x_1, \dots, x_n$ and $u_1, \dots, u_n$ are two transcendence bases of the same rational field, any identity that is invariant under field isomorphism must have the same meaning in both systems. The formalism of universal domains is designed to express statements in this coordinate-free way, while retaining enough coordinates for actual calculation.

The passage from polynomial rings to quotient fields must be made with care. In the polynomial ring, divisibility and specialization are visible. In the quotient field, every non-zero polynomial has become invertible, and exceptional cases have disappeared. A proof that uses quotient fields may therefore prove only a generic statement. To recover a statement about specializations one must return to a localized polynomial ring and record which denominators have been inverted.

Algebraic adjunctions are handled by prime ideals in polynomial rings over the transcendental base. If an element y is to satisfy an irreducible equation $F(y) = 0$, the universal domain is the quotient by the prime ideal generated by the chosen irreducible factor. If the equation factors after extending constants, the different factors correspond to different branches. This is the algebraic origin of the distinction between absolute and relative irreducibility.

The independence theorem for transcendence bases underlies the entire construction. If K is finitely generated over k , every maximal algebraically independent subset has the same cardinality. Furthermore K is algebraic over the rational field generated by such a subset. Thus the entire domain can be built in two stages: a completely transcendental stage and

an algebraic stage. This two-stage decomposition is the model for many later normalization arguments.

The most general domain for a problem is therefore not unique as a ring with named generators, but it is unique up to the evident universal property. Any other domain satisfying the same formal relations receives a specialization from it, after excluding denominators. This universal property is what justifies calculations with generic coefficients and generic roots. The generic calculation is a calculation in an initial algebraic situation from which special cases are obtained by homomorphism.

Further proof details

The terminology completely transcendental is meant to exclude hidden algebraic relations. If a calculation with parameters produces a polynomial relation among them, then the parameters were not completely transcendental for that problem. The universal domain is constructed so that every relation is visible in the defining prime ideal. This is the safest setting for proofs by generic specialization.

One can also use these domains to compare different problems. If two universal domains have isomorphic quotient fields over the base field, then the two generic situations are birationally equivalent. If the isomorphism preserves the distinguished subrings, the equivalence is stronger and respects specialization. Thus the language of domains distinguishes birational equivalence from isomorphism of chosen models.

The later ideal theory will replace many of these arguments by statements about noetherian rings and integral closure. Still, the guiding idea remains the same: every general algebraic situation should be represented by a finitely generated domain whose quotient field is the field of the generic case. Once this is done, questions about specialization, dependence, and algebraic extension can be treated uniformly.

10. Functional equations of the isomorphic mapping

Original publication: 1916.

Let two algebraic domains be given by generators and relations. An isomorphic mapping between them is determined by the images of the generators, but those images must satisfy the same relations. Thus the problem of isomorphism can be written as a system of functional equations. The aim is to express this system algebraically and to determine when its solutions form a finite or rational family.

Let

$$R = k[x_1, \dots, x_n]/\mathfrak{p}, \quad R' = k[y_1, \dots, y_m]/\mathfrak{q}$$

be integral domains. A homomorphism $R \rightarrow R'$ is defined by choosing elements $X_i(y) \in R'$ such that every relation in $\mathfrak{p}$ vanishes after substitution $x_i \mapsto X_i(y)$. If the homomorphism is to be an isomorphism, there must also be inverse expressions $Y_j(x)$ satisfying the corresponding relations and the two systems of composition equations.

The functional equations therefore consist of three parts: preservation of the defining relations of R , preservation of the defining relations of R' , and inversion equations. Written with undetermined coefficients for the functions X_i and Y_j , these become polynomial equations in those coefficients. Thus the set of isomorphisms is itself represented by an algebraic system whenever degree bounds are imposed.

Degree bounds are not arbitrary. In many problems they follow from dimension, from the degree of the defining equations, or from the requirement that certain divisor classes be preserved. Once a bound is known, the existence of an isomorphism is an elimination problem. One eliminates the coefficients of the inverse functions and obtains conditions on the original domains. Conversely, a solution of the coefficient equations gives an actual isomorphism after checking the denominators.

The method applies particularly well to rational function fields. An automorphism of $k(x)$ is a fractional linear transformation, and the functional equations reduce to the invertibility of

a two-by-two determinant. In several variables the Cremona transformations appear. They are again described by polynomial equations among the coefficients of rational functions and their inverses. The complexity lies not in the concept of isomorphism but in obtaining useful degree bounds.

If a group is prescribed to act by isomorphisms, the functional equations acquire additional composition relations. For each group element S one introduces a system of functions X_i^S . The equations express

$$X^{ST} = X^S \circ X^T, \quad X^1 = X.$$

Together with the defining equations of the domain, these are the algebraic equations for a representation of the group by automorphisms. This viewpoint anticipates the treatment of equations with prescribed group.

The paper also makes clear why isomorphism is a relative notion. If the ground field is enlarged, new solutions of the functional equations may appear. An isomorphism over $\bar{k}$ need not descend to k . The descent condition is the invariance of the solution under the Galois group, possibly after conjugation by automorphisms of the object. Thus the functional equations describe both the geometric isomorphisms and the arithmetic obstruction to their definition over the ground field.

In modern notation one would say that the isomorphisms form a functor. The paper treats the same idea by direct equations. For every coefficient domain A one may ask for solutions with coefficients in A ; substitution and composition are polynomial operations, hence compatible with homomorphisms of A . The representing equations therefore encode the whole family of isomorphisms, not only the individual transformations over an algebraically closed field.

The conclusion is that the problem of recognizing isomorphic algebraic structures can be brought under elimination theory once the transformations are bounded. The functional equations are explicit, functorial, and stable under specialization. In this form the theory provides a bridge between rational substitutions, automorphism groups, and the Galois-theoretic construction of equations.

Additional reductions and formulation

One should distinguish an isomorphism from a correspondence that is generically one-to-one. A rational map may have an inverse on dense open subsets but fail to be a regular isomorphism on the chosen affine models. The functional equations detect the birational inverse, while additional denominator conditions detect where the inverse is defined. This distinction is why the equations are naturally written in function fields first and only afterward translated into coordinate rings.

If the domains carry extra structure, such as a distinguished subfield, a grading, or a divisor at infinity, the functional equations acquire preservation conditions. For example, a graded isomorphism must send homogeneous elements of degree one to homogeneous elements of degree one. Such restrictions often provide the degree bounds that are otherwise missing. The general method is therefore flexible: one writes equations not only for multiplication but also for every structure required to be preserved.

Composition of two candidate isomorphisms is polynomial in the coefficients after clearing denominators. The equations expressing that the composition is the identity are therefore algebraic equations. This is an important finiteness point: invertibility is not an infinite condition once a degree bound has been imposed. It is represented by finitely many coefficient equations obtained by comparing the numerator and denominator of each composed coordinate.

For automorphism groups, the same equations define a group law on the solution set. The identity solution and inverse solution are given by polynomial operations in the coefficients, again after localization by the determinant or resultant that expresses non-degeneracy. Thus the automorphisms form an algebraic group or an algebraic family in the cases where the degree is bounded. The later language of group schemes is not used, but the construction is already present in equation form.

The descent problem can be solved in favorable cases by averaging only after the isomorphism equations have been linearized. Direct averaging of rational transformations usually fails

because denominators do not average well. Instead one studies the induced action on a finite-dimensional vector space of forms of bounded degree. If an invariant point satisfying the non-degeneracy condition exists in that space, it gives a descended isomorphism. This is the algebraic content behind many classical descent arguments.

Further proof details

The same functional-equation method applies to equivalence rather than isomorphism. If objects are considered equivalent after multiplying equations by units, changing coordinates, or adding redundant generators, the equations are enlarged by the corresponding auxiliary unknowns. Equivalence is then expressed by the solvability of this larger system. This is how classification problems enter the same algebraic framework.

A point of the construction is that automorphisms are not first found one by one. They are found as the solution set of a finite algebraic system. This makes it possible to speak of specializations in which the automorphism group jumps. The generic automorphism group may be smaller than that of a special member, and the jump is detected by additional solutions of the functional equations after specialization.

The inversion equations are usually the most restrictive part. A homomorphism of coordinate rings is easy to write; proving that it has an inverse forces relations among the coefficients of the proposed images. In rational form these relations include non-vanishing of determinants or resultants. The algebraic equations describe the closed part of the condition; the non-vanishing conditions describe the open part.

11. Equations with prescribed group

Original publication: 1918.

Let G be a finite group. The problem is to construct equations whose group is G , and to

do so in a form that separates the formal group action from accidental specializations. This is the inverse problem in its algebraic form. One begins with a rational function field on which G acts faithfully, forms the fixed field, and then regards the original field as a Galois extension of that fixed field.

Let x_g be independent variables indexed by the elements of G , and let G act by left translation,

$$S(x_g) = x_{Sg}.$$

The field $K = k(x_g : g \in G)$ is then a regular representation field for G . Its fixed field K^G has the property that K/K^G is a Galois extension with group G . A primitive element $z \in K$ gives an equation over K^G whose roots are the conjugates $S(z)$. This equation has group G in the generic sense.

The difficulty is to replace the many variables by as few parameters as possible and to decide when specialization preserves the group. The coefficients of the generic equation lie in the fixed field. If this fixed field is rational, the equation can be written with independent parameters. If it is not known to be rational, one still obtains an equation over the fixed field, but not necessarily over a purely transcendental parameter field. Thus the rationality problem for fixed fields enters directly into the inverse Galois problem.

Specialization is controlled by the discriminant and by the resolvents corresponding to proper subgroups. If the discriminant is non-zero, the specialized equation is separable. If no resolvent belonging to a proper subgroup acquires a rational root, the group does not drop into that subgroup. Since there are only finitely many proper subgroups, the exceptional specializations are contained in a finite union of proper algebraic sets. Hence the general specialization retains the prescribed group.

The construction is invariant under change of primitive element. If z and z' both generate K over K^G , their equations have the same splitting field and the same group. The coefficients are different functions on the same parameter domain. Thus the group belongs to the extension, not to the particular polynomial. This distinction is essential when comparing equations

obtained by different resolvent constructions.

The regular representation may be replaced by any faithful representation of G by substitutions of rational functions. If the action on a rational field $k(t_1, \dots, t_n)$ is faithful, then the extension over the fixed field again has group G . A smaller representation can yield equations with fewer parameters. But the fixed field may become harder to identify. The trade-off is between the number of parameters and the rationality of the invariant field.

If G is a subgroup of the symmetric group S_n , one can take the roots themselves as variables. The elementary symmetric functions give the full symmetric fixed field, and additional resolvents distinguish G from its overgroups. This is the classical approach. The field-theoretic formulation shows that it is a special case of the same invariant construction, with the roots providing a permutation representation.

The theorem may be summarized as follows. For every finite group G there is a generic Galois extension with group G , obtained from a rational function field carrying the regular action of G . Its specializations, outside a proper exceptional set and subject to the usual separability condition, give equations whose group is contained in G ; outside the additional resolvent loci the group is exactly G . The remaining problem is to make the parameter field rational and minimal.

This conclusion shifts the inverse problem from the existence of a formal equation to the rationality and descent of invariant fields. It explains why the problem is easy to formulate for every finite group but difficult to solve over a prescribed ground field with few parameters. The group action is elementary; the arithmetic of the fixed field is the subtle part.

Additional reductions and formulation

The generic equation obtained from the regular representation is universal in a weak but useful sense. Any Galois extension with group G can be obtained by specializing the independent variables to a normal basis, at least after adjoining the necessary constants and avoiding the

discriminant. The coefficients of the generic equation then specialize to the coefficients of the given equation. This explains why the regular representation is the natural starting point despite its many variables.

Resolvents measure subgroup containment. If $H \subset G$, choose a rational function fixed by H and by no larger subgroup. Its conjugates under G give a resolvent equation whose rational root after specialization is equivalent to the Galois group lying in a conjugate of H . Thus the group is exactly G if none of the resolvents for maximal proper subgroups has a rational root. This is the classical resolvent criterion in invariant-field language.

The construction also shows why different polynomials may have the same group but different parameter fields. Changing the primitive element changes the embedding of the extension into a polynomial equation. The splitting field and its group remain the same, but the coefficients may generate a smaller or larger subfield of the invariant field. The best equation is often the one whose coefficients generate the smallest rational subfield over which the extension is still regular.

When G has a faithful linear representation of small dimension, one may replace the regular variables by coordinates in that representation. If the invariant field is rational, this gives a generic equation with fewer parameters. The obstruction is exactly the rationality problem for $k(V)^G$. Thus Noether's problem on rationality of invariant fields appears not as a side issue but as the central compression problem for equations with prescribed group.

The exceptional specialization set is not merely the discriminant. The discriminant prevents collisions among roots; resolvent conditions prevent the group from falling. There can also be denominator conditions coming from the chosen primitive element and from the chosen rational generators of the fixed field. The correct statement is therefore always: outside a finite union of proper algebraic exceptional sets, the specialization has the prescribed group.

Further proof details

For a subgroup chain $H \subset G$, resolvents can be arranged compatibly. A root of the H -resolvent gives a reduction of the group to a conjugate of H ; further resolvents then test smaller subgroups. In this way the lattice of subgroups is reflected in a tower of resolvent equations. The generic construction contains all these resolvents at once, because every intermediate fixed field of K/K^G corresponds to a subgroup of G .

The field of constants remains a serious issue. Over an algebraically closed field the regular representation construction is formal. Over a smaller field, the desired group action and the primitive element must be defined over that field. If the group representation requires roots of unity or other constants, the resulting equation over the smaller field may have a different arithmetic behavior. This is another form of the descent problem.

The final shape of the theorem is therefore twofold. Universally, every finite group occurs as the group of a regular extension of a suitable invariant field. Arithmetically, the production of equations over a specified rational parameter field depends on rationality and descent. The distinction prevents the formal construction from being mistaken for a complete solution of the inverse Galois problem over a given base.

12. Invariants of arbitrary differential expressions

Original publication: Nachrichten der Gesellschaft der Wissenschaften zu Goettingen, 1918.

Let a differential expression be formed from dependent variables and their derivatives up to some finite order. A group of transformations of the independent and dependent variables acts not only on the variables themselves but on all derivatives, by the prolonged transformation. An invariant differential expression is one that remains unchanged under this prolonged action. The purpose is to describe such invariants without restricting to special forms of the expression.

If $x = (x_1, \dots, x_n)$ are independent variables and $u = (u_1, \dots, u_m)$ dependent variables, then derivatives are denoted by u_j^α , where α is a multi-index. A finite-order differential expression is a function of finitely many such jet variables. A transformation group acts on the jet variables by the chain rule. The invariance condition is therefore a system of differential equations for the expression.

For an infinitesimal group with generators

$$X_\nu = \sum_i \xi_{\nu i} \frac{\partial}{\partial x_i} + \sum_j \eta_{\nu j} \frac{\partial}{\partial u_j},$$

the prolonged generators $X_\nu^{(r)}$ act on derivatives up to order r . A differential expression I of order r is invariant precisely when

$$X_\nu^{(r)} I = 0$$

for all generators. Thus the determination of invariants becomes the integration of a complete system of first-order linear partial differential equations on jet space.

The completeness of this system follows from the closure relations of the infinitesimal group. If $[X_\mu, X_\nu]$ is a linear combination of the generators, then the same is true after prolongation. Consequently the equations $X_\nu^{(r)} I = 0$ are compatible. Their solutions are functions of a maximal set of independent differential invariants. The number of such invariants is the dimension of the jet space minus the rank of the prolonged group action.

Invariant differentiation is the operation that carries invariants to new invariants. An ordinary derivative of an invariant need not be invariant, because the independent variables themselves transform. One must form special differential operators

$$D_1, \dots, D_s$$

which commute with the prolonged group action up to linear combinations. Applying these operators to known invariants yields further invariants. This is the differential analogue of

forming covariants from algebraic invariants by transvection.

The finiteness question is more delicate than in polynomial invariant theory, since differentiation produces infinitely many expressions. The appropriate statement is not that there are only finitely many invariants, but that all invariants can be generated from finitely many fundamental differential invariants by invariant differentiation and algebraic operations. The proof depends on the stabilization of the ranks of the prolonged actions and on the ability to solve for higher derivatives in terms of invariant derivatives.

The treatment is independent of the special nature of the differential expression. It applies to scalar functions, systems of functions, and expressions involving several independent variables. What changes is the jet space and the rank of the prolonged group. The method remains the same: prolong the infinitesimal transformations, solve the invariant equations, and construct invariant differentiations.

A useful normal form is obtained by choosing a cross-section to the group orbits in a sufficiently high jet space. The coordinates of the transformed jet at the cross-section are differential invariants. Once the cross-section is fixed, every other invariant is a function of these normalized quantities and their invariant derivatives. This normal form is algebraic when the group action is algebraic and the cross-section is chosen by algebraic equations.

The conclusion is that the invariant theory of differential expressions is structurally parallel to the invariant theory of algebraic forms. Infinitesimal transformations replace symbolic letters; prolongation replaces polarization; invariant differentiation replaces transvection. The principal difference is that the generating process is infinite in order, though finite at each stage and controlled by finitely many fundamental operations.

Additional reductions and formulation

The prolongation formula is obtained by differentiating the transformed dependent variables and applying the chain rule. For a first derivative one obtains terms involving derivatives of

the infinitesimal coefficients ξ and η ; for higher derivatives the formula is iterated. Although the expressions become long, they are determined recursively. This recursion is what makes the invariant equations finite at every prescribed order.

The rank of the prolonged group action may increase with the order of derivatives. At low order the group may have stabilizers; at sufficiently high order the action may become free on a dense open set. The number of fundamental differential invariants at each order is then the increase in the dimension of jet space minus the increase in orbit dimension. This count is the differential analogue of the dimension count by Hilbert series in algebraic invariant theory.

Invariant differentiation is constructed by finding differential operators that transform covariantly. If $\omega_1, \dots, \omega_s$ is an invariant coframe, the dual total derivative operators preserve invariants. In coordinates this means solving a linear system whose coefficients are the transformation coefficients of the independent variables. The determinant of this system marks the exceptional jets where the construction fails or changes rank.

The method clarifies the role of moving frames even when that terminology is not used. A cross-section to the group orbits fixes the group parameters as functions of the jet variables. Substituting these parameters into the transformed derivatives gives normalized differential invariants. All other invariants are functions of these normalized quantities, provided one stays in the open set where the cross-section is valid.

Differential syzygies arise because invariant differentiations need not commute. Their commutators are linear combinations of the invariant differentiations with invariant coefficients. These coefficients themselves are differential invariants. Consequently, differentiating fundamental invariants produces relations as well as new invariants. A complete theory must include both the generating invariants and the commutation relations among invariant differentiations.

Further proof details

At each finite order the differential invariant problem is an ordinary invariant problem on a finite-dimensional jet space. What is new is the compatibility as the order increases. Prolongation maps from order $r + 1$ to order r must carry invariants to invariants, and invariant differentiations move in the opposite direction. The structure is therefore a graded tower rather than a single finite algebra.

The arbitrary nature of the differential expression means that the results are not tied to equations of motion or to a particular geometry. Curvature expressions, differential equations, and variational integrands are all treated by the same method. One specifies the transformation group, prolongs it, and solves for invariant functions on jets. The concrete interpretation of the resulting invariants depends on the problem, not on the general theory.

The finiteness statement should be read locally on the regular part of jet space. Where the rank of the prolonged action drops, additional invariants may appear or the chosen invariant differentiations may become singular. This is analogous to algebraic invariant theory, where orbit dimension may drop on special strata. The general theorem describes the open stratum; special strata require separate analysis.

13. Invariant variational problems

Original publication: Nachrichten der Gesellschaft der Wissenschaften zu Goettingen, 1918, pp. 235–257.

The problem is to determine the consequences for the Euler equations when a variational integral is invariant under a continuous group of transformations. Let

$$I = \int f(x, u, \partial u, \dots) dx$$

be an integral whose integrand depends on finitely many derivatives. If the integral is invariant

under a group, then the variational derivatives of f satisfy identities. Conversely, under suitable hypotheses such identities imply invariance up to a divergence. This is the content of the two theorems.

Write the Euler expressions as

$$\psi_i = \frac{\partial f}{\partial u_i} - \sum_{\lambda} \frac{d}{dx_{\lambda}} \frac{\partial f}{\partial u_{i,\lambda}} + \sum_{\lambda,\mu} \frac{d^2}{dx_{\lambda} dx_{\mu}} \frac{\partial f}{\partial u_{i,\lambda\mu}} - \dots .$$

The first variation has the form

$$\delta f = \sum_i \psi_i \delta u_i + \text{Div } A.$$

This formula is the basis of everything that follows. Invariance means that for every infinitesimal transformation the variation of f is itself a divergence.

For a finite continuous group depending on ρ essential parameters, the infinitesimal transformations have characteristic variations

$$\bar{\delta} u_i = \sum_{\alpha=1}^{\rho} a_{\alpha} Q_{i\alpha}.$$

Substituting in the first variation and using invariance gives

$$\sum_i \psi_i Q_{i\alpha} = \text{Div } B_{\alpha}$$

for each α . Thus there are ρ divergence relations among the Euler expressions. These are conservation laws. They are proper when the parameters are constant and independent.

This is the first theorem: to every finite continuous symmetry group of the variational integral there correspond as many independent conservation laws as there are independent group parameters. The conservation laws are explicitly obtained by inserting the infinitesimal transformations into the boundary term of the first variation. No integration of the Euler equations is required. The law is an identity in the functions and their derivatives.

The second theorem concerns infinite groups depending on arbitrary functions and their derivatives. Suppose the infinitesimal transformations contain arbitrary functions $p_1(x), \dots, p_\sigma(x)$ and their derivatives up to some finite order. After integration by parts, invariance gives identities of the form

$$\sum_i \psi_i Q_{ij} - \text{Div} \sum_i \psi_i Q_{ij}^{(1)} + \text{dots} = 0$$

for $j = 1, \dots, \sigma$. These are not conservation laws but dependencies among the Euler expressions themselves and their derivatives. They hold identically, before any field equations are imposed.

This distinction explains the difference between proper conservation laws and improper ones. For finite groups, the divergence vanishes on solutions of the Euler equations and gives a conserved current. For infinite groups, the corresponding divergence is a consequence of identities among the Euler expressions; it is therefore improper in the sense that it contains no independent conservation content beyond the identities. This resolves an ambiguity that appears in mechanics and relativity.

The converse statements are equally important. If ρ independent divergence relations of the indicated form are given, then one can reconstruct a finite continuous group under which the integral is invariant, up to a divergence. If σ independent identities among the Euler expressions and their derivatives are given, then one obtains an infinite group depending on σ arbitrary functions. Thus symmetries and identities determine each other.

The proof is algebraic integration by parts. Treat the arbitrary variations and their derivatives as independent, collect coefficients, and transfer derivatives from the arbitrary functions to the Euler expressions. Boundary terms are absorbed into divergences. The independence of the arbitrary functions forces the identities. Conversely, reversing the calculation reconstructs the variation of the integrand as a divergence.

The theorem applies to classical mechanics, where time translations yield energy conservation and spatial translations yield momentum conservation. It also applies to the general theory of relativity. The invariance under arbitrary coordinate transformations is an infinite group.

Therefore the energy relations associated with it are improper conservation laws; they arise from differential identities among the gravitational field equations. This is the mathematical explanation of the special status of gravitational energy expressions.

In the language of the calculus of variations, the first theorem associates finite-dimensional symmetry with conserved divergences; the second associates gauge-type symmetry with differential identities among Euler equations. The two cases are formally similar but conceptually distinct. The first yields constants of motion on solutions. The second yields relations that make some field equations dependent on others.

The final formulation is independent of coordinates. A variational problem is invariant if the change of the integrand under the prolonged infinitesimal transformation is a total divergence. The Euler operator annihilates divergences. Therefore applying the Euler operator to the invariance condition gives the identities; applying the first variation formula gives the conservation laws. This is the complete mechanism behind the correspondence.

Additional reductions and formulation

The finite-parameter theorem includes the familiar conservation laws as special cases. Time translation gives the energy integral; translations in space give momentum; rotations give angular momentum. The theorem does not depend on the mechanical interpretation. It says that every infinitesimal transformation for which the change of the Lagrangian is a divergence produces a divergence expression that vanishes on all extremals. The conserved quantity is read directly from the boundary term of the first variation.

The arbitrary-function theorem is more subtle. If the transformation contains an arbitrary function and its derivatives, the coefficients of that arbitrary function after integration by parts must vanish identically. These vanishing expressions are the Noether identities. They imply that not all Euler equations are independent. In gauge theories this is exactly what one expects: gauge freedom means that the field variables contain redundant descriptions of the same physical state.

The converse direction is often the most powerful part of the result. Given identities among the Euler expressions, one can reconstruct transformations with arbitrary functions. This shows that dependencies among field equations are not accidental; they are the algebraic shadow of an infinite symmetry group. In particular, if a variational problem has identities of order s , the corresponding symmetry transformations involve arbitrary functions and derivatives up to a related order.

Divergence terms must be retained throughout. Two Lagrangians that differ by a divergence have the same Euler expressions, but their Noether currents differ by an identically conserved term. Therefore the theorem is naturally formulated modulo divergences. This is not a defect: the variational problem itself is unchanged by a divergence, and conservation laws are likewise determined only up to trivial divergences.

The application to general relativity was one of the motivations for the paper. Coordinate invariance is an infinite group depending on arbitrary functions. The associated energy expressions cannot be ordinary tensorial conservation laws of the same kind as those coming from finite symmetries. They are tied to identities among the gravitational equations. The theorem gives the precise mathematical reason for this phenomenon without relying on physical terminology.

In modern terms the first theorem concerns global symmetries and the second concerns gauge symmetries. But the paper's formulation is more general than this terminology. It treats finite and infinite continuous groups uniformly by counting essential constants or arbitrary functions and by applying integration by parts to the first variation. The result is a structural theorem about variational complexes, expressed in the calculus notation of the time.

Further proof details

The Euler operator is the reason divergences disappear from the field equations. Applying it to a divergence gives zero identically. Therefore any invariance statement modulo a divergence becomes, after applying the Euler operator, an identity among Euler expressions. This simple

observation is the algebraic core of the second theorem and explains why integration by parts is unavoidable in the proof.

Independence of the conservation laws or identities is measured by the independence of the infinitesimal transformations. If a linear combination of generators is trivial modulo transformations that vanish on all fields, the corresponding conservation law is trivial. Thus the counting of laws requires the group parameters to be essential. Noether states the hypotheses so that the number of laws is the number of essential parameters or arbitrary functions.

The theorem is local in the independent variables. Boundary conditions are not part of the algebraic statement. They become relevant only when one integrates the divergence over a domain and interprets the boundary term as a conserved quantity. The identities themselves hold pointwise as formal differential identities. This is why the theorem applies equally to mechanics, field theory, and purely geometric variational problems.

14. Arithmetic theory of algebraic functions of one variable

Original publication: Jahresbericht der Deutschen Mathematiker-Vereinigung 28 (1919), pp. 182–203.

The arithmetic theory of algebraic functions of one variable stands between two older theories: the theory of algebraic number fields and the geometric theory of algebraic curves. The aim is to exhibit the precise analogy and also the differences. A function field of one variable over a constant field behaves like a number field in its ideals and divisors, while its places and differentials have a geometric interpretation on a curve.

Let K/k be an algebraic function field of one variable. A prime divisor, or place, is the

valuation corresponding to an irreducible branch of the curve. Divisors are formal sums

$$\mathfrak{a} = \sum_P n_P P$$

with almost all $n_P = 0$. The principal divisors are those of the form (f) , with $f \in K^\times$. The divisor class group is the quotient of all divisors of degree zero by principal divisors. It is the function-field analogue of the ideal class group.

The analogy with number fields becomes sharp when one studies integral elements. Choosing a place at infinity, the functions regular away from infinity form a Dedekind domain. Its non-zero ideals factor uniquely into prime ideals. The divisor theory of the complete function field is obtained by adjoining the infinite primes. Thus the arithmetic of ideals and the geometry of points are two languages for the same structure.

Ramification is measured by the different. If L/K is a finite separable extension of function fields, a prime P of K decomposes into primes Q of L , with ramification indices $e(Q/P)$ and residue degrees $f(Q/P)$. The different exponent measures the failure of the trace pairing to be integral. The sum of these exponents enters the Riemann-Hurwitz formula, just as the discriminant enters the arithmetic of number fields.

The Riemann-Roch theorem is the central theorem connecting arithmetic and geometry. For a divisor D , let $L(D)$ be the vector space of functions f such that $(f) + D \geq 0$. Then

$$\ell(D) - \ell(K_0 - D) = \deg D + 1 - g,$$

where K_0 is a canonical divisor and g is the genus. In number-field language this resembles the balance between ideals and differentials; in geometric language it is the dimension theorem for linear systems.

The theory of differentials gives the most transparent bridge. A differential has orders at all places, and its divisor is canonical. Residues satisfy the global relation that the sum of all residues is zero. This is the function-field analogue of the product formula and of trace

duality. Through differentials, the canonical class and the different of an extension acquire arithmetic meaning.

The constant field must be treated carefully. Extending constants changes the set of rational places and may split prime divisors, but it does not change the geometric function field after algebraic closure. Many statements are most natural over an algebraically closed constant field; arithmetic applications require descent to the original constants. This mirrors the distinction between absolute and relative ideal theory in number fields.

The paper compares three formulations of the same theory: ideals in an integral domain of functions regular outside a prescribed divisor, divisors on the complete curve, and valuations of the function field. Each has advantages. Ideals make unique factorization of ideals immediate. Divisors make linear equivalence and Riemann-Roch natural. Valuations make extension and ramification transparent. A complete arithmetic theory must pass freely among the three.

The conclusion is that the function-field case is not a mere analogy to number fields. It is an exact arithmetic theory in which the geometric model supplies additional tools. Conversely, the number-field theory can be illuminated by translating its notions into the language of valuations, divisors, differentials, and class groups. This perspective prepares the later abstract construction of ideal theory in both number and function fields.

Additional reductions and formulation

The divisor attached to a function records zeros and poles with multiplicity. The product formula says that the total degree of a principal divisor is zero. This is the simplest sign that the theory is global. Local valuations determine the divisor, but the degree condition ties all places together. In the number-field analogy, this is the counterpart of the product formula for absolute values.

The ideal-theoretic and divisor-theoretic languages differ mainly by the treatment of infinity. If one removes a finite set of places S , the functions regular outside S form a Dedekind

domain. Ideals in this domain correspond to effective divisors away from S . Adding the places in S recovers the complete divisor group. Thus changing the set of infinite places changes the affine arithmetic but not the complete function field.

The Riemann-Roch theorem also explains why genus is an arithmetic invariant. It measures the defect in the naive expectation that a divisor of degree d should impose d independent conditions. In genus zero the function field is rational. In higher genus the failure of rational parametrization is reflected in the correction term involving the canonical divisor. This connects rationality of one-variable fields with the arithmetic of divisors.

In extensions of function fields the different has two equivalent descriptions. Locally it is defined by the trace dual of the integral closure of a valuation ring. Globally it is a divisor whose degree enters Riemann-Hurwitz. The local exponents measure ramification, while the global degree measures the change of genus. This exact correspondence is one of the strongest reasons for treating function fields and number fields by the same ideal-theoretic methods.

The constant-field issue appears in the definition of degree of a divisor. If the constant field is not algebraically closed, a place may have residue degree greater than one. Its contribution to the divisor degree is weighted by that residue degree. After extending constants, the place may split into several geometric points. The arithmetic theory over the original field must keep the residue degrees; the geometric theory over an algebraic closure sees the split points.

Further proof details

The comparison with number fields is especially close for the trace pairing. If A is the integral closure of a Dedekind domain in a finite extension, the inverse different consists of elements whose trace against A is integral. This definition works in number fields and function fields alike. In the function-field case, the divisor of the different can also be read geometrically from ramification of the corresponding curve map.

Linear equivalence of divisors is the function-field form of principal ideal equivalence. Two

divisors differ by a principal divisor exactly when their associated fractional ideals differ by multiplication by a field element, after choosing the affine ring of functions regular away from infinity. The class group of divisors is therefore the natural completion of the ideal class group. Riemann-Roch then gives information about representatives of these classes.

This perspective makes clear why one-variable function fields are the correct analogue of number fields rather than of arbitrary multivariable function fields. Their local rings at non-singular points are discrete valuation rings, and their non-zero ideals factor into primes. In higher dimension prime divisors still exist, but ideals have more complicated primary decompositions. The one-dimensional case is where the arithmetic analogy is exact.

15. Finiteness of the system of integral invariants of binary forms

Original publication: 1919.

The problem is to prove the finiteness of the system of integral invariants for binary forms. Here integral means that the invariants are polynomials with integer coefficients in the coefficients of the ground forms. The usual proofs over fields do not immediately give integral generators, since division by integers may occur. The task is to avoid such division or to control it so that the final generators remain integral.

Let a binary form of degree n be written

$$f = a_0x^n + a_1x^{n-1}y + \cdots + a_ny^n.$$

The group SL_2 acts on the variables and hence on the coefficients. An integral invariant is a polynomial $I(a_0, \dots, a_n) \in \mathbb{Z}[a_0, \dots, a_n]$ fixed by this action. For several binary forms one takes the diagonal action on all coefficient sets. The invariant ring over $\mathbb{Z}$ is the object to be generated.

Over a field of characteristic zero, Hilbert's theorem gives finite generation. But the proof uses averaging or differential operators that may introduce rational coefficients. One may clear denominators for any one invariant, but this does not by itself produce a finite integral generating system. The integral theorem requires showing that all denominators needed in the field proof can be absorbed into finitely many integral forms.

The method is to compare the invariant ring over $\mathbb{Z}$ with its extensions to $\mathbb{Q}$ and to the residue fields modulo primes. Let $R_{\mathbb{Z}}$ be the integral invariant ring and $R_{\mathbb{Q}}$ its rational extension. Choose rational generators of $R_{\mathbb{Q}}$, clear denominators, and obtain finitely many integral invariants. They generate a subring $A \subset R_{\mathbb{Z}}$ such that $R_{\mathbb{Z}}/A$ is torsion. The remaining question is to control this torsion uniformly.

Reduction modulo a prime p detects the possible missing integral invariants. If an invariant is not generated by A , its residue modulo some prime gives a new invariant in characteristic p . Therefore it is enough to prove finite generation in every characteristic in a manner compatible with lifting. For binary forms the representation theory supplies enough operators to reduce high-degree invariants to lower-degree ones, even modulo p , after separating the exceptional primes.

The symbolic method again plays a role. Transvectants of binary forms can be written with integral coefficients if the binomial factors are normalized correctly. The apparent divisions in classical transvectants are artifacts of convention. By choosing integral symbolic operators, one obtains covariants and invariants whose coefficients lie in $\mathbb{Z}$. Products and contractions of these integral covariants remain integral.

A key lemma states that, beyond a computable degree, every invariant is congruent modulo the ideal generated by positive-degree invariants of lower degree to an invariant of a restricted normal form. The number of such normal forms is finite in each degree range. Combined with the rational finite-generation theorem, this gives a bound beyond which no new integral generator is required. The argument is a graded version of the usual ascent from quotient modules.

The theorem then follows from the noetherian property of the polynomial ring over $\mathbb{Z}$. The invariant ring is a graded subring, and the reductions show that it is integral over the finitely generated subring A up to bounded torsion. Since only finitely many primes and finitely many degrees remain exceptional, adjoining finitely many further integral invariants completes the generating system.

The conclusion is that the full ring of integral invariants of any finite system of binary forms is finitely generated over $\mathbb{Z}$. This is stronger than finite generation over $\mathbb{Q}$: it asserts that a single finite list of invariant polynomials with integer coefficients generates all integral invariants without introducing denominators. The result is an integral refinement of Hilbert's finiteness theorem.

Additional reductions and formulation

One begins with a graded invariant ring. The degree-zero part is $\mathbb{Z}$, and the positive-degree part is an ideal. If a finite set of homogeneous invariants generates this ideal up to a given degree and if every higher-degree invariant is reducible modulo lower degrees, the whole ring is generated by the finite set. The proof is the standard induction on degree, but the difficulty is to make the reduction integral.

The passage from rational to integral invariants can be organized prime by prime. For a prime p , reduce the coefficient ring modulo p and consider invariants over $\mathbb{F}_p$. If the reductions of the chosen integral generators generate the invariant ring in characteristic p up to the required bound, then no new generator with denominator divisible by p is needed. Only finitely many primes can fail the chosen finite checks.

Binary forms are special because the SL_2 -operation is controlled by two fundamental differential operators, raising and lowering the weight. In characteristic zero this gives complete reducibility. Integrally, the operators must be normalized to avoid denominators. The resulting congruence arguments show that high-weight terms can be reduced to transvectants and products already in the generated ring.

The theorem does not assert that the integral generating system is minimal. Minimality over $\mathbb{Z}$ is not stable under reduction modulo primes, and generators that are redundant over $\mathbb{Q}$ may be needed integrally to account for congruence phenomena. The statement is finite generation: there exists one finite list that suffices for all integral invariants. This is the right analogue of Hilbert's theorem over the integers.

The result is important because it shows that invariant theory over the integers is not merely rational invariant theory with denominators cleared case by case. It is a noetherian theory in its own right. Integral invariants form a finitely generated algebra, and arithmetic questions about congruences of invariants can be studied inside that algebra without returning to infinitely many separate denominators.

Further proof details

The proof also requires that products of invariants behave well under reduction. If an invariant reduces modulo p to a product of lower-degree invariants, then after lifting the factors one obtains the original invariant modulo p and lower-degree terms. Repeating this process and using induction on degree reduces the integral problem to finitely many congruence classes. This is the arithmetic form of the reduction argument.

The exceptional primes are unavoidable because representation theory of SL_2 changes in characteristic p . Operators that are invertible over $\mathbb{Q}$ may acquire kernels modulo p , and new invariants may appear in small characteristic. The theorem allows this: one adds finitely many integral lifts to handle the exceptional primes. What it excludes is the need for infinitely many new generators as the degree grows.

The final integral generating system can be chosen homogeneous. This follows from the grading by degree in the coefficients. If a non-homogeneous invariant is present, each homogeneous component is invariant as well, because the group action preserves degree. Homogeneous generators are therefore sufficient and make the induction on degree possible.

17. Modules in non-commutative domains, especially from differential and difference expressions

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The theory of modules over commutative polynomial rings had already shown the power of ideal-theoretic methods. The purpose here is to extend part of that theory to non-commutative domains that arise naturally from differential and difference operators. In such domains multiplication depends on commutation rules, and left ideals need not be right ideals. One must therefore formulate every statement with attention to side.

The basic examples are operator rings. In a differential operator ring one has a coefficient field K and an operator D satisfying

$$Da = aD + D(a)$$

for $a \in K$. In a difference operator ring one has an automorphism S and a rule

$$Sa = a^S S.$$

Both rings are non-commutative unless the derivation or automorphism is trivial. Nevertheless they admit division algorithms in suitable cases and therefore possess ideal-theoretic structure.

A left module over such a ring represents a system of linear differential or difference equations. Generators correspond to unknown functions; relations correspond to equations. Transforming the system by elementary operations is the same as replacing the module presentation by an equivalent presentation. Thus module theory gives an invariant language for systems of operators.

The first question is the chain condition. A domain is left-Noetherian if every ascending chain of left ideals stabilizes. For operator rings of the above type, the proof uses an order filtration. The leading term of an operator is commutative enough to be compared by degree.

An ascending chain of left ideals gives an ascending chain of leading coefficient ideals or leading monomial sets, which stabilizes. One then descends on order to show that the original chain stabilizes.

Because left and right multiplication differ, one must also study right ideals separately. In many operator rings the same filtration proves both left and right chain conditions, but the generators and reductions are not identical. The distinction becomes important when forming quotient modules and when asking for common right or left multiples of operators. The classical least common multiple of differential operators is a one-sided concept.

The elementary divisor theorem does not carry over unchanged. Over a principal ideal domain every finitely generated module decomposes into cyclic modules. In non-commutative domains, cyclic decompositions may fail or may depend on side. What remains is a theory of presentations, syzygies, and reductions to normal form under elementary row and column operations that respect the non-commutative multiplication rules.

For differential equations, this means that a system can be simplified by eliminating unknowns and equations, but the resulting single equation, when it exists, is not canonical in the same way as in the commutative case. The annihilator of a module and the common multiples of the operators encode the invariant content. The module itself is the correct object; a chosen scalar equation is only one representation of it.

The paper develops divisibility notions adapted to non-commutative rings. If $a = bc$, then b is a left divisor and c a right divisor. Greatest common divisors and least common multiples must be qualified accordingly. Euclidean arguments, when valid, produce one-sided greatest common divisors. These are sufficient for many transformations of differential and difference systems.

The conclusion is that a substantial part of ideal and module theory survives in non-commutative operator domains once one respects the side conditions and uses filtrations. This opens the way to treating differential and difference equations by algebraic methods analogous to the ideal theory of polynomial equations. Later non-commutative algebra will

make these ideas systematic, but the essential module viewpoint is already present here.

Additional reductions and formulation

The order filtration is the central technical tool. Assign to an operator its order in D or S . The leading coefficient of a product is the product of the leading coefficients, with the appropriate automorphism in the difference case. Therefore leading terms behave sufficiently like commutative monomials to permit induction. Lower-order commutator terms do not destroy the argument because they are handled after the leading term has been reduced.

Syzygies must also be one-sided. A relation among left module generators has coefficients acting from the left. If the same symbols are moved to the right, the relation changes. This is not a notational inconvenience but an algebraic fact. For systems of differential equations it corresponds to the distinction between combining equations by applying operators to them and combining unknown functions by operator substitutions.

A finitely generated left module over a left-Noetherian operator ring has a finite presentation. This means that the relations among a finite set of equations are themselves finitely generated. In analytic language, compatibility conditions of a linear differential system can be generated from finitely many identities. The algebraic theorem does not solve the equations, but it gives a finite algebraic description of their formal consequences.

The comparison with commutative elementary divisors shows both what survives and what fails. Torsion modules still make sense: an element is torsion if some non-zero operator annihilates it. But annihilators are left ideals, and their factorization is not controlled by commutative prime ideals. Decomposition theory must therefore be replaced by ideal chains, filtrations, and one-sided divisibility.

The examples from difference equations are parallel to the differential ones but reveal the role of automorphisms more clearly. The rule $Sa = a^S S$ means that coefficients are shifted when an operator passes them. Common multiples correspond to bringing equations to a

common shift range. The algebraic operations are the same as for differential operators after replacing derivation by automorphism in the commutation rule.

The paper's broader message is methodological. Non-commutative algebra should not be treated as a collection of exceptional calculi. Operator domains have ideals, modules, chain conditions, quotient modules, and presentations, provided the definitions are made on the correct side. This is the point of departure for a structural theory of non-commutative rings analogous to the ideal theory of commutative domains.

Further proof details

The filtration also gives an associated graded ring. In the differential case the associated graded ring is often commutative or at least simpler than the original operator ring. Ideals and modules can then be studied by passing to leading terms in the associated graded ring and lifting reductions back to the filtered ring. This is the method behind the chain-condition proofs.

For systems of equations, finite presentation has a concrete computational meaning. Starting from finitely many equations, there are finitely many generating compatibility conditions; compatibility conditions among those are again finitely generated, and so on in each finite stage allowed by the ring's homological behavior. The paper does not use modern homological language, but it already treats equations through their modules of relations.

Non-commutativity forces care with localization. In a commutative domain one may invert a multiplicative set without choosing sides. In an operator domain, denominators must satisfy compatibility conditions if a ring of fractions is to exist. The differential and difference examples often have enough Ore-type behavior for the necessary fractions, but this is a theorem, not a convention. The module theory must respect this limitation.